

White-Matter Hyperintensity (WMH) Segmentation

Week 1 Deliverable

BME 4408 — Medical Imaging Course Project

Dataset: MICCAI 2017 WMH Segmentation Challenge found at
<https://doi.org/10.34894/AECRSD>

What is WMH and FLAIR??

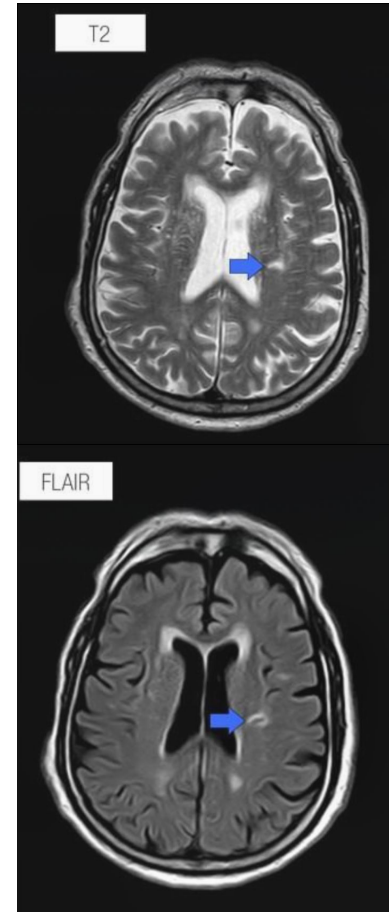
WMH - White Matter Hyperintensity

WMHs are areas of the brain's white matter that appear unusually bright on certain MRI scans. They indicate changes or abnormalities in white-matter tissue and can be associated with factors such as aging, neurological conditions, etc.

FLAIR - Fluid-Attenuated Inversion Recovery

FLAIR is an MRI sequence designed to suppress the signal from cerebrospinal fluid (CSF). It works using an RF pulse followed by a carefully timed 180° refocusing pulse. The timing is chosen so that the longitudinal magnetization of CSF crosses zero when the image is acquired, effectively cancelling its signal.

In T2 scans, the CSF turns out bright and makes it quite difficult to distinguish it from surrounding tissues. So FLAIR was invented to zero out CSF, and make tissue abnormalities that were obscured by bright CSF more visible.



Sample Images

This dataset was collected at three hospitals, each on a different MRI scanner:

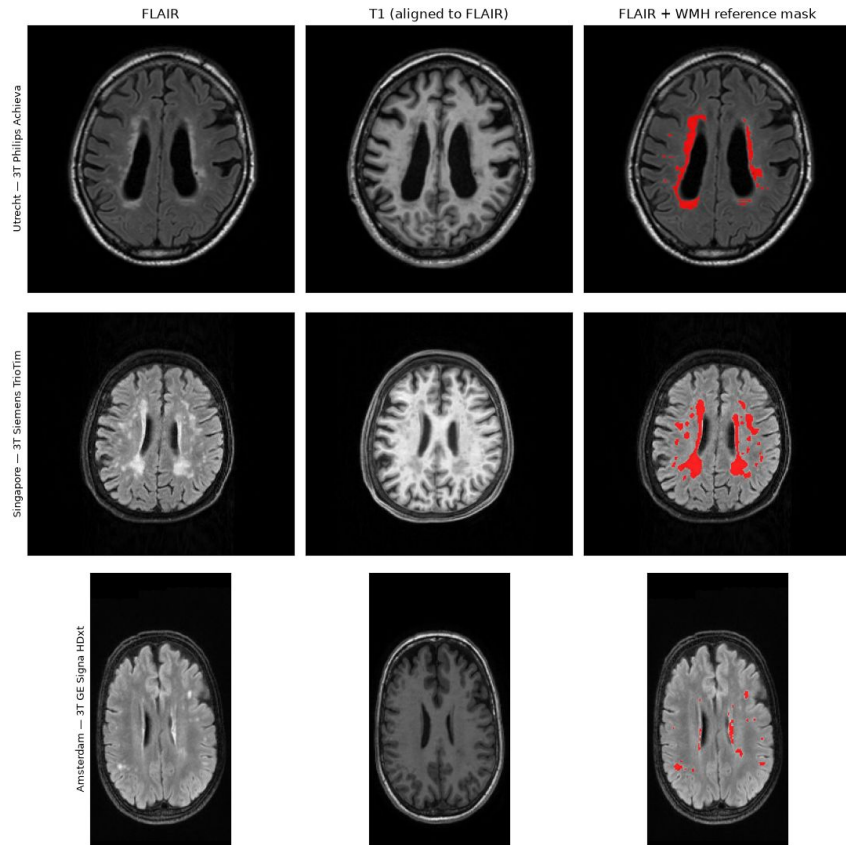
- Utrecht: 3T Philips Achieva
- Singapore: 3T Siemens TrioTim
- Amsterdam: 3T GE Signa HDxt

Here is one training subject from each site.

- T1
- FLAIR
- FLAIR overlaid with the WMH reference mask.

Even at a glance you can see contrast and slice-thickness differences between scanners.

That cross-scanner variability is a real theme we'll have to handle later.



Data Distribution

Our problem is **segmentation**, not classification.

The subjects are distributed across:

- a) Sites/scanners
- b) How much WMH lesion burden each subject carries
- c) Voxel-level imbalance between WMH and background within a single scan.

Patient demographics Subjects were **70.1 ± 9.3 years old, 50% male** (Kuijf et al., 2019).

a) Sites/Scanners

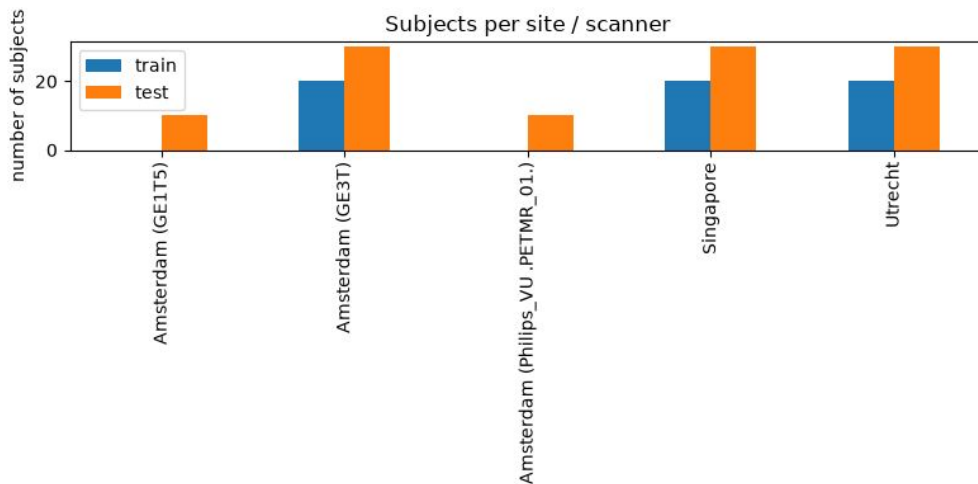
Two of the five test scanners never appear in training at all.

These are unseen scanners that the eventual segmentation model will be tested against.

Unlike a 2D chest X-ray dataset, each subject here isn't one image. It's a set of 3D brain **volumes**:

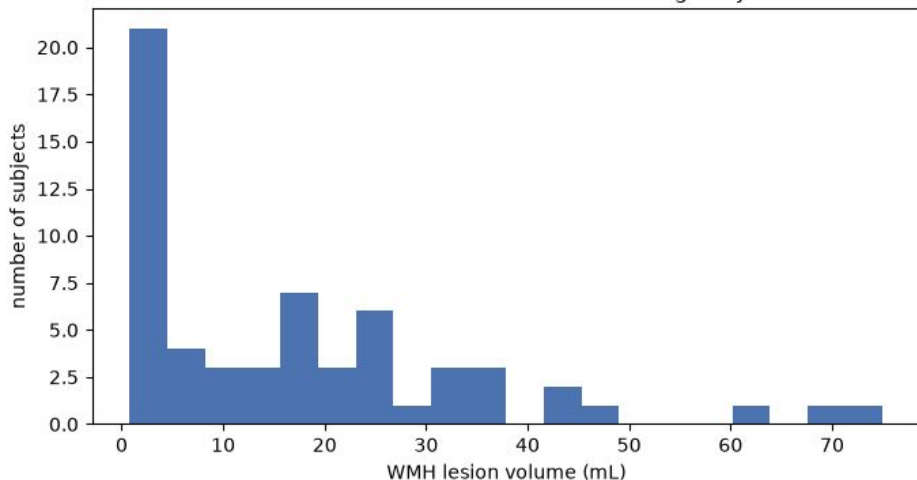
- an original of both T1 and FLAIR
- a bias-field-corrected version of both
- plus the WMH reference mask

	train	test
Amsterdam (GE1T5)	0	10
Amsterdam (GE3T)	20	30
Amsterdam (Philips_VU .PETMR_01.)	0	10
Singapore	20	30
Utrecht	20	30
TOTAL	60	110



b) How much WMH lesion burden each subject carries

Lesion burden distribution — 60 training subjects

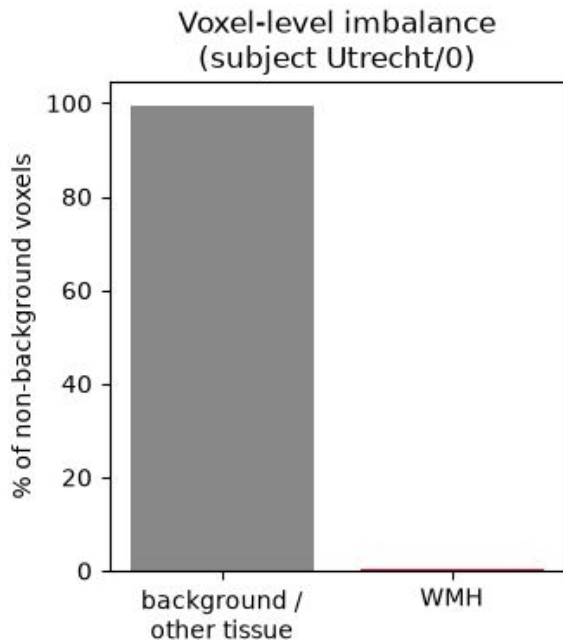


```
count    60.000000
mean     17.610413
std      17.374935
min       0.786000
25%       3.090353
50%      15.058497
75%      25.905071
max      74.991382
Name: lesion_volume_ml, dtype: float64
```

A long right-hand tail (a few subjects carry a very heavy lesion load) and nobody in this cohort has zero lesions. That skew is exactly why any later severity classification (mild/moderate/severe) should work on the log-transformed volume rather than the raw mL value.

c) Voxel level imbalance

This is the imbalance that actually matters for a segmentation model. What fraction of image voxels are WMH vs normal tissue?



```
non-zero FLAIR voxels (rough head proxy): 1,738,050  
WMH voxels: 8,778  
WMH share: 0.505 %
```

The File Format

What is NIfTI?

NIfTI (.nii / .nii.gz) is the standard file format for 3D (and 4D) brain imaging data in research pipelines.

One file = one whole 3D volume, not dozens of separate 2D slice files the way DICOM stores a scan.

A NIfTI file has two parts:

1. A small header (metadata)
2. The image array (voxel intensities).

The header carries an **affine matrix** which maps every voxel index (i, j, k) to a real-world position in millimetres.

This is what lets software know exactly where and how big a voxel is.

```
file: training/Utrecht/0/pre/FLAIR.nii.gz
array shape: (240, 240, 48)
voxel spacing (mm): (0.958, 0.958, 3.0)
data type: float32
orientation (axis codes): ('L', 'P', 'S')

affine matrix (voxel index -> world mm):
[[-9.55749035e-01 -4.81770411e-02 1.60402849e-01 1.18610664e+02]
 [ 2.20411010e-02 -8.68188620e-01 -1.26836812e+00 1.27181854e+02]
 [ 6.67887405e-02 -4.02902126e-01 2.71394706e+00 1.33168182e+01]
 [ 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.00000000e+00]]
```

The File Format

Why NIfTI and not DICOM?

Scans come off the scanner as DICOM (one file per 2D slice)

For a research pipeline, DICOM is not ideal. A single brain scan can be 50–300+ separate files, slice ordering has to be reconstructed, etc. so it's a huge hassle.

NIfTI packages the whole 3D volume as **one array + one affine** which is exactly what volumetric operations need.

The dataset organisers already performed the DICOM → NIfTI conversion during acquisition.

Reading and Displaying the Data in Python

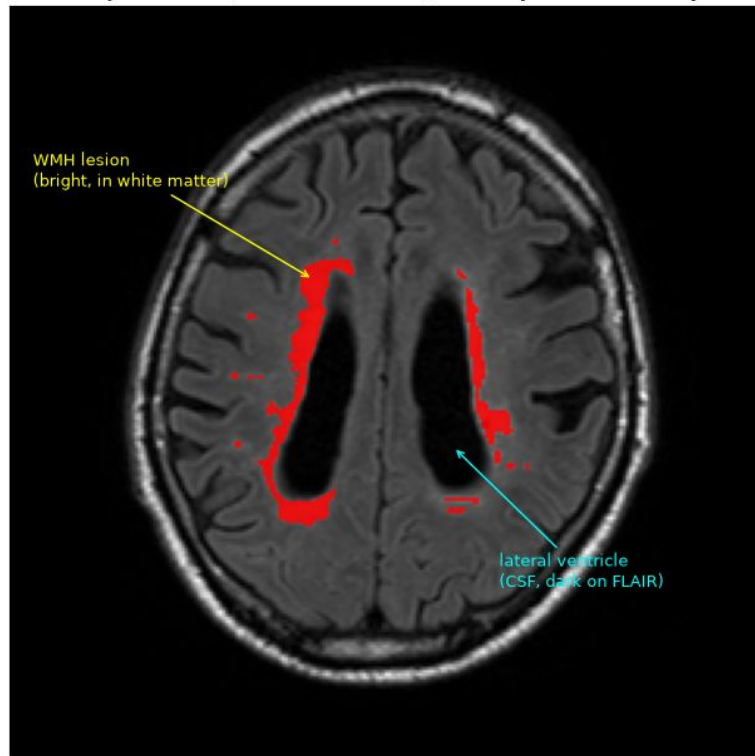
The anatomical structures relevant to this project are labelled directly on the image here.

- The ventricles
- WMH lesion

Ventricles are dark since FLAIR is specifically designed to suppress CSF signal.

Lesions are bright, sitting in the white matter. They are red in this image because of the WMH overlay.

Subject Utrecht/0 — axial slice 33, anatomy + WMH overlay



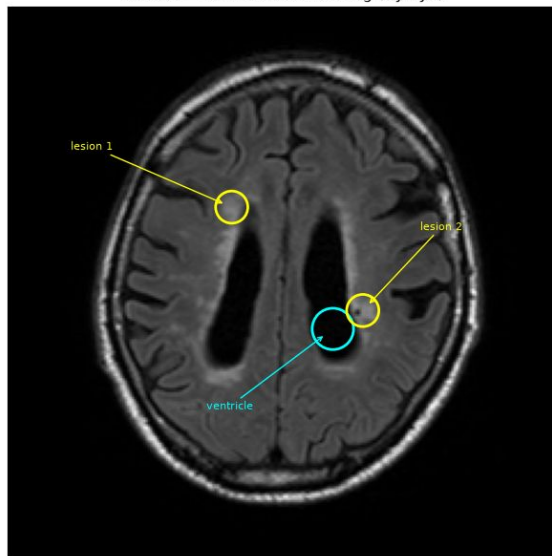
Manually Marking the Segments of Interest

Hand-markings done on a few sample images for the structures we'll later identify with code.

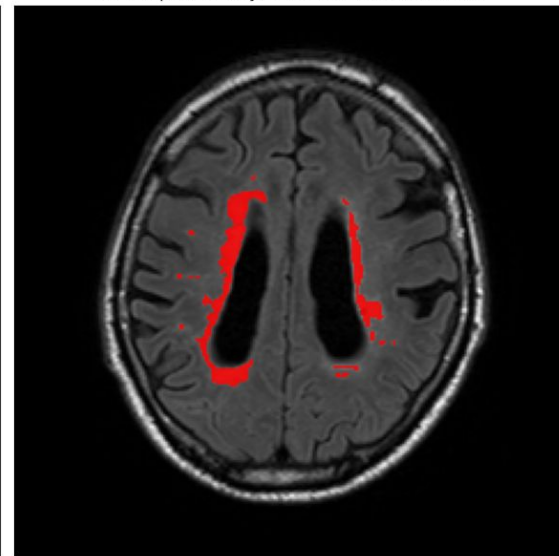
Panel A: Circling the ventricle and the two clearest WMH lesions on this slice by eye.

Panel B: The expert-drawn reference mask that ships with this dataset (wmh.nii.gz), shown purely as a reference for comparison.

Panel A — our manual marking (by eye)



Panel B — expert reference standard (wmh.nii.gz)
for comparison only, not what we hand-marked



Project Objectives

This project designs a pipeline to detect and segment WMHs, clinically tied to cerebrovascular disease, ageing, and dementia risk.

In plain terms, here's what we're building:

1. **Pre-process** each scan
2. **Segment** WMH lesions on FLAIR
3. **Label** each lesion anatomically (periventricular vs. deep white matter)
4. **Extract quantitative features** per subject
5. **Classify** overall WMH lesion as mild/moderate/severe, and benchmark the whole pipeline against the official WMH Segmentation Challenge leaderboard so our results mean something outside our own dataset.

The Team

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Thank You